

Eduardo Venegas Hernandez

AI & ML Engineer | Specialized in Agentic AI, RAG Systems, Data Engineering and MLOps.
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SKILLS

AI & Data: GenAI, Agentic AI, PyTorch, LangGraph, LangChain, RAG, PySpark, SQL, Knowledge Graphs, Neo4j, ETL Pipelines
Software & Frameworks: Python, C/C++, TypeScript/JavaScript, FastAPI, Flask, Node.js, React, Next.js, Angular, Streamlit
Cloud & MLOps: AWS, Azure, OCI, Kubernetes, Docker, MLflow, CI/CD, Linux, Grafana, MQTT, PLCs

WORK EXPERIENCE

Data & AI Engineer Masterarbeit BMW & German Research Center for Artificial Intelligence Munich, Germany 03/2026 - 09/2026

- Topic: *Evaluating Multi-Turn Agentic RAG Systems: A Synthetic Benchmark Generation Approach.*
- Designed a scalable evaluation framework for failure localization and performance analysis of tool-using **LLM agents**.
- Developed a **LangGraph**-based **synthetic benchmark generation pipeline** for evaluating multi-turn agent trajectories.
- Built an unsupervised auditing framework to assess and scale **RAG** deployments across **4M+ corporate records**.

AI Research Engineer Werkstudent August-Wilhelm Scheer Institute Saarbrücken, Germany 10/2024 - 02/2026

- Industrialized multiple AI systems spanning **KG RAG**, **Agentic AI**, **Digital Product Passports**, and **Industry 4.0** applications.
- Built AI pipelines for information extraction, reasoning, and anomaly detection across **distributed devices**.
- Designed multi-agent workflows incorporating **graph traversal**, **tool calling**, **retrieval**, and self-correction mechanisms.

Big Data Platform Engineer II Oracle Guadalajara, Mexico 11/2023 - 10/2024

- Engineered **CI/CD frameworks** and deployment tooling for distributed **OCI** big data services.
- Built automated testing and release-validation infrastructure to improve reliability across cloud platform deployments.

Data Science & Analytics Intern BMW Group Munich, Germany 03/2023 - 09/2023

- Developed an **anomaly detection model** to identify vehicle battery depletion from **large-scale vehicle telemetry**.
- Engineered **PySpark** cloud services, increasing service demands by **62%** while supporting recall investigations.
- Built **NLP analytics tools**, and interactive dashboards to support model optimization and operational decision-making.

AI & DevOps Engineer Xira Mexico City, Mexico 01/2022 - 04/2022

- Managed distributed **DevOps infrastructure**, implementing automated monitoring, testing, and deployment workflows.
- Developed a Docker **API management platform** integrating Stripe payments, authentication, and service observability.

Production Engineer Fellow Major League Hacking & Meta New York, USA 06/2021 - 09/2021

- Completed fellowship at Facebook focused on **Linux administration**, **DevOps**, **SRE**, and scalable infrastructure.
- Automated deployments on **AWS EC2** using Docker and CI/CD while monitoring with **Prometheus** and **Grafana**.

Machine Learning & Full Stack Engineer Jiit Technologies Guadalajara, Mexico 10/2020 - 06/2021

- Developed demand forecasting models and data pipelines using **Python** (Flask, SciPy, Prophet, and SQL).
- Built an ERP platform using **Angular** to manage inventory, logistics, and financial operations across business workflows.

EDUCATION

MSc. Data Science and Artificial Intelligence Universität des Saarlandes Saarbrücken, Germany 10/2024 - 09/2026

- **Thesis (Seminar Grade 1.0):** Evaluating **Multi-Turn Agentic RAG Systems** (In collaboration with **BMW & DFKI**).
- Focus areas: *Machine Learning, NLP, Machine Translation, Computer Vision, Statistics, Quantum AI and ML Cybersecurity.*

BSc. Computer Science Tecnológico de Monterrey Guadalajara, Mexico 08/2019 - 12/2023

- GPA: **94.6/100**; awarded a **60% merit-based scholarship** and recognized with the **Best GPA** distinction five times.
- Focus areas: *Machine Learning, Time Series, Algorithms & Optimization, Cloud Computing, Distributed Systems, and PLCs.*

DAAD Exchange Scholar - Computer Science Universität Ulm Ulm, Germany 10/2022 - 09/2023

- Awarded a full **DAAD scholarship** for academic exchange studies in Germany.
- Focus areas: *Computer Vision, Autonomous Driving, Driver Assistance Systems, Human-Vehicle Interaction, and Statistics.*

Associate of Science (A.S.) in Robotics and Automation Engineering CETI Colomos Guadalajara, Mexico 08/2015 - 06/2019

- Focus areas: *Industrial Robotics, PLCs, Control Systems, Digital Electronics, Sensors, Instrumentation, and Automation.*

PROJECTS

Enterprise Knowledge Graph RAG Platform AWSi

- Developed an enterprise **KG-RAG system** over technical documentation using **LangGraph**, **LangChain**, **Neo4j** and **Docker**.
- Implemented **multi-turn reasoning agents** combining BFS traversal, pathfinding and self-correcting retrieval.

VectorDrive-Route: Autonomous BEV Trajectory Planning Github

- Built a **PyTorch** pipeline using **nuScenes**, **ResNet** and **depth lifting** to map multi-camera feeds into a **Bird's-Eye-View grid**.
- Designed a temporal **motion planning head** with **drivable-area** and **occupancy decoders** to generate safe paths.

Daten-Mapping-Framework für die AAS DOI: 10.17560/atp.v68i5.2836

- Developed a framework to transform enterprise data into **Asset Administration Shell** models for **Industry 4.0** interoperability.
- Designed a dual mapping strategy supporting **hierarchical asset structures** for automated schema alignment.

CERTIFICATES

- **AWS Certified Solutions Architect - Associate (SAA-C03)** Expected - Oct 2026
- **Mathematics for Data Science Specialization** HSE - 2021
- **Data Science Professional Certification** Harvard - 2020
- **Deep Learning Professional Certification** IBM - 2020

LANGUAGES

- **Spanish:** Native
- **English:** C2
- **German:** C1